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Faculty of Engineering, Design and ICT

**An AI Powered Advisory Board Agent for Corporate
Governance:**

**A Retrieval Augmented Agentic System with Explainable
Financial Reasoning**

Final Project Thesis

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for the degree of MSc in Artificial Intelligence*

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Declaration of Authorship

I, Hatem Isa, declare that this thesis titled "An AI Powered Advisory Board Agent for Corporate Governance: A Retrieval Augmented Agentic System with Explainable Financial Reasoning" and the work presented within it are my own. I confirm that:

- This work was carried out wholly while in candidature for the MSc in Artificial Intelligence at Bahrain Polytechnic.
- Where any part of this thesis has previously been submitted for a degree or any other qualification, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given.
- This thesis is entirely my own individual work, and where I have drawn on external tools this is disclosed in the AI Usage Statement.
- I have acknowledged all main sources of help.
- I have disclosed the use of any generative AI tools in the AI Usage Statement.

Signed: Hatem Isa

Date: 24 August 2026

Abstract

Governance runs on reading. A board overseeing a portfolio meets filings of several hundred pages apiece, every reporting cycle, and the question answering tools on offer handle them one question at a time, in isolation, leaving behind no trail that anyone could audit. CyberBoard-AI was built against that frustration.

It is an advisory board agent, and its parts are ordinary enough on their own. Hybrid retrieval fuses BM25 sparse scoring with dense embeddings, a cross encoder reranks whatever comes back, and a ReAct style agent above both picks between three purpose-built tools: a financial summarizer, a risk identifier, and a compliance checker. Every answer names its sources and states a confidence. The corpus is 116 United States Securities and Exchange Commission filings from 20 companies, cut into 18,136 chunks, with Bahrain Bourse annual reports added for a second market, the Gulf Cooperation Council.

Development turned up a defect worth reporting. Ticker filtering was running after global candidate selection instead of before it, which quietly threw away a company's own passages. Fixing that one ordering error moved the FinanceBench retrieval hit rate from 54.8 percent to 96.8 percent. Reranking's measured contribution climbed too, from 3.2 to 12.9 percentage points.

End to end numbers come from a hand verified set of 35 Gulf Cooperation Council questions. Over five runs the full agent averaged 96.8 percent answer accuracy and 99.4 percent faithfulness. Single pass retrieval, answering the same questions, reached 82.9 percent. Strip retrieval away entirely and the closed book model manages 11.4 percent. A paired McNemar test on the primary run found the agent strictly better wherever the two diverged, five wins to zero losses, at an exact p value of 0.0625, a figure that says more about how small the evaluation set is than about how weak the effect is.

So the contribution is not any single component. It is the assembly: retrieval, agentic tool use, and explainability brought together for corporate governance advisory, then measured across a United States benchmark and Gulf Cooperation Council filings, with every result reproducible from published result files.

Keywords: retrieval augmented generation; agentic systems; corporate governance; explainable AI; financial question answering; hybrid retrieval

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AI Usage Statement

I used Claude Code (Anthropic) as an efficiency tool for code implementation, including the Streamlit dashboard and the agent tools, under my direct supervision. I tested each generated function against actual project data before accepting it. I used Perplexity to locate candidate research papers for the literature review, and I manually retrieved and verified each paper against its original source before citing it. AI assistance was limited to code and to locating sources, all of which I critically reviewed and edited. I take full authorship responsibility for this thesis.

Table of Contents

Declaration of Authorship.....	2
Abstract.....	3
Acknowledgements.....	4
AI Usage Statement.....	5
Table of Contents.....	6
List of Figures.....	9
List of Tables.....	9
List of Abbreviations.....	10
Chapter 1 Introduction.....	11
1.1 Background and Context.....	11
1.2 Problem Statement.....	11
1.3 Motivation and Significance.....	12
1.4 Research Questions and Objectives.....	12
1.5 Contributions.....	13
1.6 Thesis Structure.....	13
Chapter 2 Literature Review.....	15
2.1 Overview of the Field.....	15
2.2 Foundations: Language Models and Financial Text.....	15
2.3 Retrieval Augmented Generation for Financial and Legal Documents.....	16
2.4 Agentic Architectures and Tool Use.....	18
2.5 Hybrid Retrieval Infrastructure.....	19
2.6 Evaluation, Faithfulness, and Financial NLP Benchmarks.....	20
2.7 Explainability and Corporate Governance.....	21
2.8 Comparative Analysis of State of the Art.....	23
2.9 Research Gap.....	23
2.10 Summary.....	24

Chapter 3 Methodology.....	26
3.1 Research Design.....	26
3.2 System Architecture.....	26
3.3 Datasets.....	28
3.3.1 Sources and Licensing.....	28
3.3.2 Pre-processing.....	28
3.3.3 Data Splits.....	28
3.4 Models and Algorithms.....	29
3.4.1 Baseline Models.....	29
3.4.2 Proposed Approach.....	29
3.5 Implementation Stack.....	30
3.6 Evaluation Methodology.....	30
3.6.1 Metrics.....	30
3.6.2 Baselines and Statistical Testing.....	31
3.7 Ethical Considerations.....	31
Chapter 4 Implementation.....	33
4.1 Development Environment.....	33
4.2 Data Pipeline.....	33
4.3 Model Implementation.....	34
4.4 Indexing and Embedding Procedure.....	34
4.5 Software Engineering Practices.....	35
4.6 Challenges and Solutions.....	35
4.7 Explainability and the Reasoning Trace.....	36
Chapter 5 Results and Evaluation.....	40
5.1 Experimental Setup.....	40
5.2 Primary Results.....	40
5.3 Ablation Studies.....	42

5.4 Error Analysis.....	43
5.5 Statistical Significance.....	44
5.6 Computational Cost and Environmental Impact.....	45
Chapter 6 Discussion.....	46
6.1 Interpretation of Findings.....	46
6.2 Comparison with Prior Work.....	46
6.3 Limitations.....	47
6.4 Responsible AI Reflection.....	48
6.5 Practical Implications.....	49
Chapter 7 Conclusion and Future Work.....	51
7.1 Summary of Contributions.....	51
7.2 Answers to the Research Questions.....	51
7.3 Future Work.....	52
7.4 Concluding Remarks.....	52
References.....	54
Appendix A Source Code Repository and Reproduction.....	57
Appendix B Dataset Details.....	57
Appendix C Hyperparameters.....	58
Appendix D Ethics Approval.....	58
Appendix E Supervisor Meeting Log.....	58

List of Figures

Figure 1. <i>CyberBoard-AI system architecture, showing the offline indexing stage and the online query and reasoning stage.</i>	27
Figure 2. <i>The CyberBoard-AI advisory dashboard in Ask Agent mode. A board member asks a question in natural language and the agent returns a grounded, source attributed answer.</i>	37
Figure 3. <i>The CyberBoard-AI reasoning trace for the query cybersecurity risk factors, showing candidate scoring, hybrid fusion, and cross encoder reranking, with source attribution and a confidence score on each result.</i>	38
Figure 4. <i>The evaluation view of the dashboard, rendering the FinanceBench retrieval hit rate and the per company breakdown from the stored result files.</i>	39
Figure 5. <i>Answer accuracy across the three configurations on both corpora, the Gulf Cooperation Council set and the recent United States set. The agent bar is the mean over five runs, matching Table 3; the two baselines are a single run on the same questions. Source: the baseline result files and the multi run summaries.</i>	41
Figure 6. <i>Retrieval hit rate by configuration on the Gulf Cooperation Council corpus.</i>	43

List of Tables

Table 1. <i>Research objectives O1 to O5.</i>	13
Table 2. <i>Comparison of representative approaches and where this thesis sits.</i>	23
Table 3. <i>End to end answer quality on the 35 question Gulf Cooperation Council set. The full agent figures are the mean over five runs; the two baselines are a single run on the same questions. Source: data/gcc_eval/multi_run_summary.json and the two baseline result files.</i>	41
Table 4. <i>Retrieval ablation on the United States corpus (FinanceBench). Source: data/financebench/ablation_results.json.</i>	42
Table 5. <i>Retrieval ablation on the Gulf Cooperation Council corpus. Source: data/gcc_eval/ablation_results.json.</i>	42
Table 6. <i>Dataset summary.</i>	57
Table 7. <i>Key hyperparameters.</i>	58

List of Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Matching 25, a probabilistic sparse retrieval function
CBB	Central Bank of Bahrain
GCC	Gulf Cooperation Council
HNSW	Hierarchical Navigable Small World, an approximate nearest neighbour index
LLM	Large Language Model
MD&A	Management Discussion and Analysis
NLP	Natural Language Processing
RAG	Retrieval Augmented Generation
ReAct	Reasoning and Acting, an agent prompting pattern
RQ	Research Question
SEC	United States Securities and Exchange Commission
XAI	Explainable Artificial Intelligence

Chapter 1 Introduction

1.1 Background and Context

Oversight is a duty that comes down, in practice, to reading. A single 10-K filed with the United States Securities and Exchange Commission can pass several hundred pages across five regulated sections, business overview through controls and procedures, and a board following a portfolio absorbs many of them each reporting cycle. The material is dense, shaped tightly by regulation, and consequential: a missed risk factor, a misread control statement, and what follows can be a governance failure.

Expectations in the Gulf Cooperation Council run along similar lines. The Central Bank of Bahrain Corporate Governance Code, to take the relevant example, assigns board responsibilities that quietly assume a working command of the filings beneath them.

Large language models and retrieval augmented generation have lately made it possible to put questions to long documents in plain language. The tools that reach practitioners are another matter. They take each question on its own, without reasoning across documents, without surfacing risk unprompted, and without the traceable evidence a board would want before trusting any answer at all. This thesis goes after that gap. What it builds is an advisory board agent that retrieves from corporate filings, reasons over what it retrieves through a small set of purpose built tools and returns answers carrying both source attribution and a confidence score.

Two gaps sit behind that. The first is evidential. A tool that answers without showing which passage it answered from asks the reader to take fluency as proof, and in a governance setting the reader has no way to discharge that trust. The second is geographic. Almost all this work is built and tested on United States filings, so how it behaves on Gulf Cooperation Council disclosure is not a settled question but an open one, and a regional institution reading both markets has no evidence to go on.

1.2 Problem Statement

Artificial intelligence tools for financial documents have not yet moved decisively past passive question answering. Structured reasoning across several documents is uncommon. Tools that surface risks unprompted, or ground every claim in a cited source, are rarer still. On the evidence of the review in Chapter 2, no published system combines retrieval augmented retrieval with agentic multi step reasoning for corporate governance advisory and

then evaluates that combination on both a United States benchmark and Gulf Cooperation Council filings. That absence is the problem this thesis takes on.

1.3 Motivation and Significance

The motivation behind it is practical and narrow. A board member has little use for a chat interface producing fluent prose. What is wanted is an answer checkable against the filing it came from, with a stated confidence and a clear boundary drawn around what the system knows. Significance follows from exactly that. In a governance setting, a system that cites its sources and abstains when the evidence is missing is safer than one answering confidently out of parametric memory. Evaluating on Gulf Cooperation Council filings next to a United States benchmark tests something further, which is whether an approach developed against one market carries to another regulatory context. That question is what decides regional applicability.

A second reason emerged from the evaluation rather than motivating it at the start. A model answering from memory alone reaches 63.2 percent on the recent United States questions and 11.4 percent on the Gulf Cooperation Council ones, because the disclosure of very large United States issuers is well represented in the text such models are trained on and Bahrain Bourse annual reports are not. Retrieval grounding is therefore not an equal improvement everywhere. It matters most exactly where a market is thinly represented, which is the case for the region this thesis is written in. Section 6.2 develops the point with the measurements behind it.

1.4 Research Questions and Objectives

The thesis is organised around three research questions, unchanged from the approved proposal.

- RQ1: How effectively can an agentic retrieval augmented system retrieve and combine information from corporate filings to support board level decisions, measured by retrieval precision and answer faithfulness.
- RQ2: Does a multi tool agent architecture produce higher answer quality, greater reasoning depth, and fewer hallucinations than a single pass through retrieval augmented generation.
- RQ3: How do features such as source attribution and reasoning chain visualisation affect the transparency and verifiability of advisory outputs.

These questions map onto five objectives.

#	Objective
O1	Build a retrieval augmented pipeline that ingests, chunks, embeds, and indexes SEC and Bahrain Bourse filings.
O2	Implement a ReAct style agent with three financial tools: a financial summarizer, a risk identifier, and a compliance checker.
O3	Evaluate retrieval and answer quality on FinanceBench and on a custom Gulf Cooperation Council set.
O4	Conduct an ablation study over retrieval components with statistical analysis.
O5	Deliver explainability through source attribution, confidence scores, and reasoning traces.

Table 1. Research objectives O1 to O5.

1.5 Contributions

This thesis makes four contributions.

- The first is a single system unifying hybrid retrieval, agentic tool use, and explainability for corporate governance advisory, instead of leaving the three as separate research strands.
- The second is a cross market evaluation, one system measured on a United States benchmark and on hand verified Gulf Cooperation Council filings, every reported result reproducible from a published result file.
- The third is a documented retrieval defect and its correction, which took the FinanceBench retrieval hit rate from 54.8 percent to 96.8 percent. It is offered as a transferable lesson: test end to end behaviour, not units alone.
- The fourth is an empirical finding about where retrieval grounding matters most. Run on both corpora, the closed book baseline answers 63.2 percent of the recent United States questions from parametric memory and 11.4 percent of the Gulf Cooperation Council questions. That fivefold gap carries a methodological consequence, which is that any retrieval system evaluated on United States filings alone will understate what retrieval contributes. Section 6.2 develops the point.

1.6 Thesis Structure

The remaining chapters run as follows. Chapter 2 reviews the literature across retrieval augmented generation, agentic architectures, hybrid retrieval infrastructure, evaluation and

faithfulness, and explainability, then states the research gap. Chapter 3 sets out the methodology, covering system architecture, datasets, models, implementation stack, evaluation methodology, and ethical considerations. Chapter 4 describes the implementation, along with the engineering challenges met on the way. Chapter 5 reports results and evaluation. Chapter 6 discusses what the findings mean against prior work, where the limitations sit, and how the system should responsibly be used. Chapter 7 concludes and sets out future work.

Chapter 2 Literature Review

2.1 Overview of the Field

Five strands of research meet in the system this thesis proposes, and they matured largely in parallel: the language modelling foundations that made document grounded question answering possible, retrieval augmented generation, agentic architectures that let a model plan and call tools, the retrieval infrastructure that fuses sparse and dense signals, and the evaluation and explainability methods that establish whether an answer can be trusted. This chapter reads each strand critically, not descriptively. For every area it asks what the cited work established, but also where that work runs out, where its authors disagree, and how the limitation shaped a design decision in CyberBoard-AI. The point is to place the system inside existing knowledge rather than to dress its components up as inventions, and to reach a research gap that is specific and defensible instead of a general appeal to novelty.

Why organise thematically? Because the contribution here is integrative. Not one component is new. The claim is that a particular combination has never been assembled and evaluated for corporate governance advisory across two markets, and a thematic structure makes that argument visible, letting each mature component be judged on its own terms before the gap between them is named in Section 2.9.

Every source reviewed was found through targeted search, then confirmed individually against its original publication before citation, so that any claim attributed to a source traces back to a document actually read rather than to a summary of it. Coverage runs from the foundational architecture and financial text analysis of Section 2.2, through the retrieval and agentic method literature of Section 2.3 and Section 2.4, the retrieval infrastructure of Section 2.5, the evaluation and faithfulness literature of Section 2.6, and the explainability and Gulf Cooperation Council governance sources of Section 2.7, the regulatory code included as a primary source. Where a body of work is large, the review picks the sources bearing most directly on a design decision here rather than attempting exhaustive coverage.

2.2 Foundations: Language Models and Financial Text

Technically, everything discussed in this chapter rests on the transformer architecture of Vaswani et al. [31], which put self attention in place of recurrence and made it practical to model long range dependencies across documents as long as corporate filings. Scaling that architecture produced the general purpose capability documented in the GPT-4 technical

report [33]. That report shows strong performance on professional and academic benchmarks, and it is candid about something else: the model can state confident, fluent falsehoods. The candour matters here. Fluency without grounding is exactly the failure a governance advisory tool cannot tolerate, and it is why this design grounds every answer in retrieved evidence rather than in parametric memory.

Financial text, though, has a longer and more specific history in natural language processing than the recent large language model literature alone would suggest. Loughran and McDonald [32] showed that general purpose sentiment dictionaries misclassify financial disclosure. Words such as liability, tax, or cost carry no negative sentiment in a regulatory filing, whatever a general lexicon says, and in response they built finance specific word lists. Their work matters here twice over. It establishes that financial language is a distinct register rewarding domain aware handling, which supports the decision to segment filings by their regulatory sections instead of treating them as undifferentiated text. It also marks the start of an arc running from dictionary methods, through learned embeddings, to the retrieval grounded language models used in this thesis, and it is against that arc that the present system should be understood.

Length is the practical problem that architecture creates. Self attention compares every token with every other token, so cost climbs quadratically with sequence length, and an annual filing runs far past what any single context window handles comfortably. Chunking is the standard answer, and it is not a neutral one. Cut a document into pieces and you have already decided what counts as a retrievable unit of evidence. Section 3.3.2 returns to that decision with a specific chunk size and overlap, and the reasoning there follows directly from the constraint named here.

2.3 Retrieval Augmented Generation for Financial and Legal Documents

Retrieval augmented generation starts with Lewis et al. [1]. Their method pairs a language model's parametric knowledge with non parametric retrieval from an external store.

That pairing answers the grounding problem raised in Section 2.2 directly. It hands the model an evidence source it can cite.

Gao et al. [2] later sorted the field into naive, advanced, and modular paradigms. The survey is valuable here for its criticism of the naive form.

Retrieve once. Concatenate into the prompt. What comes back is often topically related yet does not answer the question, and there is no route to recovery when that first retrieval goes badly. The reranking and agentic layers this thesis adds come straight out of that critique. They are what Gao et al. file under advanced and modular.

Inside finance, FinSage from Wang et al. [3] builds a two phase multi path search with reranking for filing question answering, and reports high recall. Lithgow-Serrano et al. [4] survey financial documentation more broadly. Their finding is that retrieval augmented performance is uneven, sensitive above all to how evidence gets chunked and matched.

What this thesis builds is deliberately plainer than FinSage. One fused retrieval path. One reranking stage. Chapter 5 is where that simplicity gets tested.

Corroboration also arrives from outside finance. Wiratunga et al. [5] take a case based approach, and a 2026 Vietnamese legal study [6] takes another, and both show sparse plus dense hybrid retrieval with reranking to be well established in heavily regulated, document rich domains structurally close to corporate governance.

Transfer from legal text to governance text is not automatic. Structure differs, vocabulary differs. What the two share is the evidentiary demand, since an answer has to point at a specific clause, and that shared demand is what makes this architectural choice reasoned rather than arbitrary.

Chunking is exactly where Lithgow-Serrano et al. [4] locate the variance. Their review reports that retrieval augmented performance on financial documentation swings on how evidence is segmented and matched, more than on the choice of generator sitting downstream. That finding carries weight for this design. It says the retrieval layer deserves the engineering attention, and it predicts, correctly as it turned out, that a defect in retrieval would dominate any tuning of the answering model. Chapter 4 documents precisely such a defect, and Chapter 5 measures what fixing it was worth.

There is a second lesson buried in the financial retrieval work. Filings are not homogeneous prose. A risk factors section reads nothing like a management discussion, and a footnote to the accounts reads like neither. Section aware segmentation follows from that observation, and it is why identifiers in this system encode the filing section alongside the company.

2.4 Agentic Architectures and Tool Use

Yao et al. [7] gave the field ReAct, which interleaves reasoning traces with actions. Alternating between thinking and acting improves task accuracy over pure prompting. It also leaves the reasoning inspectable, which bears on the explainability goal in RQ3.

Toolformer, from Schick et al. [8], adds something complementary. A model can learn when to call an external tool without being explicitly programmed to.

Neither result is unqualified, though. An early mistake in a ReAct style loop propagates through every step after it, and each extra reasoning step costs latency and money. Chapter 5 meets that tension head on, where the agent is more accurate than single pass retrieval and also slower.

Step back, and the surveys of Wang et al. [9] and Hu et al. [10] place ReAct style tool calling at one point in a much larger design space. That space runs all the way to multi agent routing.

Tool to Agent Retrieval [11] and MasRouter [12] are the examples, each dispatching a query among many specialised agents. Capability climbs with such designs. So does opacity, because the routing decision becomes its own unexplainable step, and evaluation grows harder alongside it.

A single ReAct agent with three fixed tools is what this thesis adopts instead. That is a scoping decision made in the open, not a limitation quietly buried. It keeps the system evaluable and its decisions traceable inside a thirteen week MSc thesis. RQ2 exists to ask whether even this simpler multi tool design beats single pass retrieval, before anyone reaches for elaborate routing.

A second reason favours the simpler design. The agentic literature is much stronger on capability demonstrations than on controlled comparison.

Plenty of the work cited above shows an agent can accomplish a task. Very little isolates how much the agentic layer actually contributes over a plain retrieval baseline, on the same questions, scored by the same judge.

Here that comparison is the central empirical question rather than an afterthought. Hence the evaluation in Chapter 5, which runs the full agent, a single pass retrieval baseline, and a closed book baseline over identical items, then reports the difference with a paired statistical test. Framing RQ2 that way follows the spirit of the surveys [9], [10], which ask for firmer evidence about when agentic complexity is warranted at all.

Cost deserves naming rather than hand waving. Every reasoning step in a ReAct loop is another model call, and the calls are sequential because each depends on the last. Latency accumulates, spend accumulates, and both scale with how many steps the agent decides to take. Chapter 5 puts numbers against that, reporting mean end to end latency for the agent beside the single pass baseline. The gap is real and it is the price of the accuracy gain. Whether the price is worth paying is a judgement a board makes, not one this thesis makes for it.

2.5 Hybrid Retrieval Infrastructure

Three technologies carry the retrieval layer. Each is well established on its own, yet to the knowledge of this review they have not been assembled before for corporate governance advisory. BM25, the sparse retrieval baseline, rests on the probabilistic relevance framework of Robertson and Zaragoza [13]; it is strong on exact term matching, a ticker or a line item name, and blind to paraphrase. Dense text embeddings come from the Siamese network training of Reimers and Gurevych [14] in Sentence-BERT; that is strong on semantic similarity but capable of returning passages topically close and factually wrong. Notice that the two failure modes are complementary. That complementarity is the whole argument for fusing them instead of picking one, and the alpha weighted fusion used here expresses the argument directly.

Inside dense retrieval sits the HNSW algorithm of Malkov and Yashunin [15], an approximate nearest neighbour index that gives up a small and bounded amount of recall for a large gain in query speed. This thesis takes that trade willingly, because interactive use demands sub second retrieval. On reranking, Pradeep et al. [16] set out practical techniques for improving cross encoders, and they are equally clear that a reranker is not free: it costs a heavier model pass and can, now and then, demote a correct passage. Chapter 5 reports exactly that behaviour on the smaller Gulf Cooperation Council corpus, where reranking works out as a small net cost to recall, a concrete instance of a tension the literature already anticipated. One further benefit followed from building sparse scoring, dense similarity, and cross encoder reranking as three inspectable stages. Each could be examined on its own, which is how the ticker filtering defect described in Chapter 4 was isolated and fixed.

The division of labour between the two model types deserves spelling out, because it is what shapes the whole retrieval stage. Sentence-BERT was motivated by a specific inefficiency. A BERT style cross encoder reads a query and a passage together, jointly, which is what makes

it accurate, and that same joint reading is what makes it unusable across a large corpus, since every candidate demands its own full model pass. Reimers and Gurevych [14] answered by training a bi encoder that maps each passage to a vector once, in advance, leaving retrieval to a cheap similarity computation.

Precision and cost pull against each other, then, and neither model type escapes the tension alone. Which is why this system uses both, in sequence. The bi encoder, paired with the HNSW index, sweeps the corpus cheaply and returns a shortlist. The cross encoder then reads that shortlist properly, query and passage together, and reorders it. Expensive accuracy gets spent only where it can change the outcome.

Two consequences follow, and Chapter 5 tests both. Widen the shortlist and the reranker has more to work with, so the size of that candidate pool becomes a parameter worth measuring rather than guessing. Narrow it too far, or truncate it accidentally, and the reranker inherits a pool that may not contain the right passage at all, in which case no amount of reordering can help. That second failure is not hypothetical. It is what Section 4.6 describes.

2.6 Evaluation, Faithfulness, and Financial NLP Benchmarks

Take the numerical reasoning benchmarks first, since they define what the field considers hard. FinQA [18] pairs questions with expert written reasoning programs over earnings report data, making the multi step arithmetic explicit rather than leaving it implicit in an answer string. TAT-QA [19] works over hybrid contexts where a table and its surrounding narrative must be read together, which mirrors how financial statements actually present information. ConvFinQA [20] extends the problem across dialogue turns, so a later question depends on reasoning established earlier in the conversation.

Each raises the bar in a different direction, and together they make one thing plain: numerical questions over filings behave differently from descriptive ones. Hence the decision here to hold the nineteen numerical questions on recent United States filings as a set of their own rather than folding them into the main evaluation. A single blended accuracy figure would have hidden whichever category was weaker.

PACIFIC [21] moves along a different axis again, asking whether a system recognises that a question is underspecified and seeks clarification instead of guessing. That capability is out of scope here, and saying so plainly is more honest than implying full coverage. A board member in this system asks a question and receives a grounded answer; the system does not

interrogate the questioner. Section 7.3 notes clarification seeking as a direction worth pursuing.

Why FinanceBench [17], then, as the primary retrieval benchmark rather than any of these? Because the question this thesis asks first is a retrieval question, not an arithmetic one. FinanceBench pairs questions with the specific evidence passages that answer them, which is exactly what is needed to score whether the right passage was found. Its limitation here is temporal: the questions cover 2016 to 2023 while this corpus covers 2024 to 2026. That mismatch is the cost of the choice, and it is why the benchmark measures retrieval quality through evidence overlap rather than answer accuracy against its own gold answers.

Turn to how faithfulness gets measured and the literature grows more self critical. It begins with the LLM as judge methodology of Zheng et al. [22], adopted here but not uncritically, since a model judge can be inconsistent or biased. Es et al. [24] built Ragas around reference free evaluation, decomposing a pipeline into faithfulness, answer relevance, and context relevance so that each can be scored without a gold answer to compare against. Useful, plainly. It also concentrates the burden onto the judge, since removing the reference removes the check on the judge as well.

Liu et al. [23] press on that weakness directly, treating judge consistency as a property to be measured rather than assumed. Song et al. [25] approach hallucination from the training side with a tuning pipeline aimed at reducing it, while Sriramanan et al. [26] go the other way, detecting hallucination from signals internal to a single response with no external knowledge required.

Set side by side, these four describe an instrument still under construction. That framing is what this thesis inherits, and it is why the judge here was treated as something to validate rather than trust. The validation paid for itself. Section 4.6 records a defect in which the judge was scoring faithfulness against a truncated view of the retrieved evidence, understating it on precisely the corpus with the longest tool outputs. An unexamined judge would have reported that number as a finding.

2.7 Explainability and Corporate Governance

Explainability in finance carries a built in conflict. Fritz-Morgenthal et al. [27] hold that a financial AI system must be explainable in two ways at once: technically accurate, and clear to a reader who is not technical. The CFA Institute report on explainable AI in finance [28] adds a second axis. Boards, regulators, and clients do not want the same explanation; each

audience needs its own from the one system. Put those together and a dilemma appears. Explain faithfully to the model internals and a board member cannot read it. Explain for readability and you risk misdescribing how the answer was really produced. This thesis takes a practical way out. Each answer arrives with the evidence under it, the source document and the section, and a calibrated confidence value beside it. The result is grounded and readable together, and it never pretends to lay bare the full internal computation.

Governance is the one place where this review reads the regulation directly instead of commentary on it. The Central Bank of Bahrain Corporate Governance Code 2022 [29] states the board responsibilities the system exists to support. The intent behind that code comes from an earlier document, Governor Al Maraj's 2009 policy rationale [30]. Why go to the primary text? The thesis makes a cross market claim on Bahrain Bourse filings, and any claim about the Gulf Cooperation Council regulatory context must stand on the regulation itself, not on a paraphrase of it.

The regulatory text is worth reading closely rather than citing in passing. The Central Bank of Bahrain code [29] frames board responsibility around informed oversight, which presumes directors can actually reach the information their oversight depends on. That presumption is where a tool like this one earns its place. A system that surfaces the relevant disclosure, names its source, and states how confident it is serves the oversight duty; a system that answers fluently from memory quietly undermines it.

Al Maraj's earlier rationale [30] supplies the intent behind the code, and intent matters when a technical design has to be justified against a regulatory expectation rather than merely aligned with its wording. Reading both together is what allows Section 6.5 to argue practical implications for Gulf Cooperation Council boards without overreaching into claims this thesis has not evidenced.

One boundary should be stated here rather than left implicit. Explainability as delivered in this system means traceability of evidence, not interpretability of the model. A board member can see which passage produced an answer, from which document and section, with what confidence, and can follow the retrieval trace that selected it. What they cannot see is why the language model composed the sentence it did. Wilson [28] would call that a partial answer for one stakeholder group, and that assessment is accepted rather than argued with. Section 6.3 carries it as a limitation.

2.8 Comparative Analysis of State of the Art

The table below sets representative systems next to each other on the dimensions this thesis cares about: how each retrieves and reasons, the domain it was built and tested for, and the metric its authors report. This is not a ranking. Scoring systems built for different tasks on one axis would be unfair, and it would miss the point. What the table shows instead is narrower and more useful. Each system pushes one or two of the dimensions this thesis needs, and leaves the others alone.

Approach	Method	Domain	Reported metric
Lewis et al. [1]	Dense retrieval plus generation	Open domain QA	Establishes RAG framework
Wang et al. FinSage [3]	Two phase multi path search and rerank	Financial filings	High recall on filing QA
Yao et al. ReAct [7]	Interleaved reasoning and acting	General agents	Improved tool use accuracy
Wiratunga et al. [5]	Case based hybrid retrieval	Legal QA	Hybrid beats single retriever
This thesis	Hybrid retrieval, ReAct agent, rerank, XAI	Corporate governance (US and GCC)	96.8% US retrieval hit; 96.8% GCC answer accuracy (5-run mean)

Table 2. Comparison of representative approaches and where this thesis sits.

One pattern runs down the rows. Retrieval systems are strong on domain and metric, then stop at retrieval. Agent systems are strong on method, but stay general purpose. The case based legal system is close in spirit, yet it is neither financial nor tested on Gulf Cooperation Council disclosure. The last row is this thesis. It carries the hybrid retrieval and reranking of the retrieval systems, the reasoning and acting loop of the agent systems, and the explainability the governance literature keeps demanding, and it reports a retrieval metric and an end to end answer metric on two markets at once. No earlier row sits at that intersection. That is the observation the next section, the research gap, makes precise.

2.9 Research Gap

Individually, retrieval augmented generation for financial documents, agentic tool use, hybrid retrieval infrastructure, and explainability in finance are each active and separately maturing research areas, and Section 2.2 to Section 2.7 have shown that each rests on solid and well

critiqued foundations. What the literature does not contain, as far as this review has found, is a single system that combines all of them into a corporate governance advisory function and evaluates that function on both a United States benchmark and Gulf Cooperation Council filings. The nearest neighbours each fall short of the combination. Financial retrieval systems such as FinSage [3] provide strong retrieval but do not add an agentic reasoning layer or evaluate on Gulf Cooperation Council disclosure. Agentic work such as ReAct [7] provides the reasoning pattern but is not specialised to financial governance or grounded in a filing corpus. Explainability work in finance [27], [28] argues for stakeholder appropriate explanation but does not deliver an evaluated end to end system. That combination, not any single component in isolation, is the specific contribution this thesis targets, and it is the gap the proposal identified and that this review confirms remains open.

Stating the gap this precisely also states what would close it, which matters for judging whether the thesis delivers. Three things are required together: a system that fuses sparse and dense retrieval with reranking over a filing corpus, an agentic layer whose contribution is measured against a matched baseline rather than asserted, and evidence surfaced in a form a non technical board reader can check. Any two of these already exist in the literature reviewed above. It is the third leg, in each case, that is missing.

2.10 Summary

The reviewed work supports every major design decision of CyberBoard-AI while leaving the integrated system unbuilt, and in several places it anticipates the exact tensions this thesis later observes: the cost of agentic reasoning against its accuracy gain, the occasional harm a reranker does to recall, and the need to validate a faithfulness judge rather than trust it. Hybrid retrieval with reranking is established, the ReAct pattern is established, LLM as judge faithfulness scoring is established, and the need for stakeholder appropriate explanation in finance is documented. What remains is to assemble these into one governance advisory system and to measure it honestly across two markets. The next chapter sets out how these pieces are combined into a single methodology.

Worth carrying forward, too, is a caution the review raises about its own field. The agentic literature reports capability far more often than it reports controlled comparison, and the evaluation literature is still arguing about whether its own instruments can be trusted. Both cautions shaped what follows. They are why the agent here is measured against matched

baselines on identical questions rather than demonstrated on examples, and why the faithfulness judge is treated as a component that could itself be wrong.

Chapter 3 Methodology

3.1 Research Design

This is a build and evaluate study. A working system answers the research questions, and its behaviour is measured against baselines and through an ablation of its own components. Four measures carry the evaluation: retrieval hit rate, answer accuracy, faithfulness, and hallucination rate. Every reported result is written to a result file in the public repository, so any figure in Chapter 5 can be recomputed from the same code and the same data rather than taken on trust.

Directness is what a build and evaluate design buys. Control is what it pays.

Because the system is constructed and then measured, the findings speak to this configuration on these corpora. Nothing licenses stretching them to a different retrieval stack, or a different market, without doing the work again.

Where it can, the design compensates. Baselines are ablations of the same system rather than separate implementations, so a comparison isolates the component under test instead of tangling it up with unrelated engineering. Same questions. Same judge. What varies between conditions is the one thing being studied.

3.2 System Architecture

CyberBoard-AI is a retrieval pipeline with an agent above it, and the split between the two carries most of the design.

The pipeline does the offline work. It downloads each filing, parses it into named sections, chunks it, embeds those chunks, and indexes them. None of that happens while a board member waits.

Query time is different. The retrieval layer fuses two signals, BM25 sparse scores and dense vector similarity, under an alpha weighted scheme; a cross encoder then reranks the fused candidates and the layer returns the top passages. Above both sits a ReAct style reasoner, choosing among three function calling tools, each built to retrieve and format evidence for one class of board level question. What the agent returns arrives wrapped the same way every time: a source attribution, a per source confidence score, and an advisory disclaimer.

Figure 1 draws this as two connected stages. The offline stage is the ingestion path: it takes filings from both corpora, parses each into named sections, chunks them recursively, embeds

them, and writes them into ChromaDB. The online stage starts with a board member's question. From there the ReAct agent picks one of the three tools, that tool queries the shared hybrid retrieval module, the module reads the index, and the agent assembles the answer, source attributed, with a confidence score and an advisory disclaimer attached.

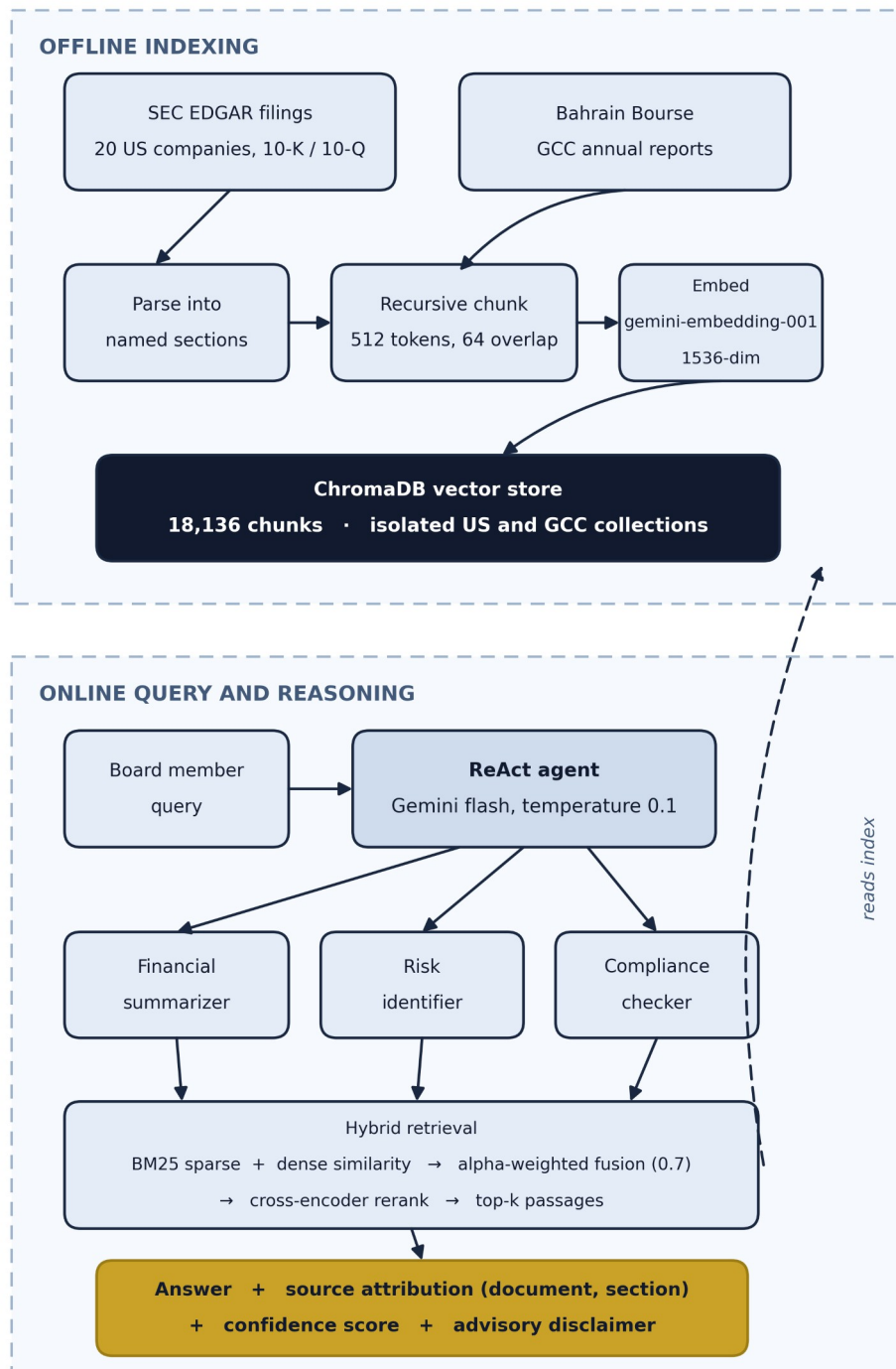


Figure 1. CyberBoard-AI system architecture, showing the offline indexing stage and the online query and reasoning stage.

3.3 Datasets

3.3.1 Sources and Licensing

The primary corpus is 116 SEC EDGAR filings, of type 10-K and 10-Q, drawn from 20 companies. Licensing raised no question here: these are public regulatory disclosures, and the ingestion script in the repository will download them again from scratch. A secondary corpus of Bahrain Bourse annual reports carries the cross market evaluation in the Gulf Cooperation Council context. FinanceBench [17] supplies the external evaluation set, its questions matched to companies in the corpus by ticker overlap.

3.3.2 Pre-processing

Before indexing, a filing is cut along its own regulatory seams. Five section categories come out: business overview, risk factors, management discussion and analysis, financial statements, and controls and procedures. Each is then split into chunks of roughly 512 tokens, with a 64 token overlap between neighbours.

The split is recursive rather than a blunt fixed length cut. Paragraph boundaries are tried first, sentence boundaries next, and word boundaries only when a single unit still runs past the token limit. Most chunks therefore end up aligned with the document's real structure instead of sliced through the middle of a sentence. For the full SEC corpus the process yields 18,136 chunks.

Both numbers are compromises, and worth defending as such.

Cut too small and a figure gets stranded from the label identifying it. In financial disclosure that is fatal, since a number means nothing without its row heading. Cut too large and the embedding dilutes: one vector asked to carry several distinct claims carries none of them sharply, and the passage wastes context when it reaches the model.

The overlap answers a narrower worry. Without it, a fact sitting on a boundary is split across two chunks and whole in neither.

3.3.3 Data Splits

Nothing here is trained, so the usual train and test split does not apply. The corpus is simply the retrieval index. The evaluation sets sit apart from it: the FinanceBench matched questions for the United States corpus, and a hand verified set of 35 questions for the Gulf Cooperation Council corpus. Those Gulf Cooperation Council gold answers earned their status the hard way. Each was checked against the source report before it was trusted as ground truth, and

three of them turned out to be wrong. The grounded agent was what flagged those three, and they were corrected during the check.

3.4 Models and Algorithms

3.4.1 Baseline Models

Two baselines isolate the value of the full system, and each strips away one thing.

The closed book model answers without retrieval, which measures how much of the task can be solved from parametric memory alone. The single pass retrieval augmented model retrieves once and answers, with no agent and no tool selection, which measures what the agentic layer adds on its own. Both are ablations of the same codebase rather than separate implementations, so what differs between any two conditions is the component removed and nothing besides.

3.4.2 Proposed Approach

At the centre is a ReAct agent [7], routed to Gemini through a LiteLLM interface and given three function calling tools to choose from. One, the financial summarizer, handles the quantitative side: revenue, earnings, and similar filing figures. A second, the risk identifier, pulls disclosed risk factors and will narrow them by category, whether regulatory, market, operational, cybersecurity, or supply chain. The third, the compliance checker, surfaces governance, internal controls, and audit disclosures, and can be pointed at a specific topic such as board oversight or environmental, social, and governance reporting. All three share one hybrid retrieval pipeline; what differs is that each scopes its query to the company in question by routing to the right collection.

Fusion weighting sits at 0.7 dense and 0.3 sparse, applied once both scores are normalised to a zero to one range so that two differently scaled signals can be added at all.

Why favour the dense side? Because most evaluation questions are put in natural language rather than in the vocabulary of the filing. The sparse component keeps a meaningful share regardless, since tickers, line item names, and defined terms are exactly where exact matching beats embedding similarity.

None of that is left as an assertion. Section 5.3 sweeps the weighting and reports what happens.

3.5 Implementation Stack

The stack is ordinary Python, chosen to be so. ChromaDB holds the vectors and provides HNSW indexing [15]. rank-bm25 does the sparse scoring. sentence-transformers runs the cross encoder reranker, ms-marco-MiniLM-L-6-v2, and Streamlit drives the dashboard. The agent itself is the OpenAI Agents SDK ReAct pattern.

Everything that touches the model goes through LiteLLM to Gemini: embeddings from gemini-embedding-001 at 1536 dimensions, generation from a Gemini flash model at temperature 0.1. Routing both through a single interface is what made the provider migration described in Section 4.6 a change of configuration rather than a rewrite of the codebase. All of it lives in a public GitHub repository alongside the README, the MIT licence, and the evaluation result files.

3.6 Evaluation Methodology

3.6.1 Metrics

Three measures do the work.

Retrieval quality is hit rate: a retrieved set counts as a hit when it holds the gold evidence. FinanceBench needed an adjustment here, because its gold spans do not line up document by document with a corpus covering 2024 to 2026, so a hit is registered whenever more than 30 percent of a gold span's words turn up in a retrieved chunk. Answer quality is simpler, just accuracy against the gold answer. Faithfulness is the subtle one, scored by an LLM judge in the manner of Zheng et al. [22]: an answer grounded in the retrieved context is faithful, and one that is not grounded is logged as a hallucination. That judge runs at temperature zero and returns structured output.

Hit rate is a blunt retrieval metric. The bluntness is deliberate.

Ranking measures such as mean reciprocal rank reward a system for putting the right passage first rather than third, which matters when a human scans a result list. That is not what happens here. The retrieved set goes to an agent that reads all of it, so the outcome turns on whether the evidence is present, not on where in the set it sits. Metric and consumption match.

That judge sees four things: the question, the gold answer, the predicted answer, and the retrieved context. It scores correctness and faithfulness as two decisions, not one.

Separating them is what makes the error analysis in Section 5.4 possible. An answer can be faithful to its evidence and still wrong, which is precisely what the single pass baseline turns out to be. Blend the two into one quality score and that distinction disappears.

3.6.2 Baselines and Statistical Testing

Pairing is what makes the comparison work. Both baselines from Section 3.4.1 meet the same judge, on the same questions, so every comparison sits paired at the question level.

Why insist on that? Because the agent and the single pass baseline answer identical items, and for identical items the correct test of an accuracy difference is McNemar on the paired outcomes. Chapter 5 reports it. Variance gets its own treatment: the evaluation runs several times over, and any run the provider cut off part way is dropped rather than folded in, so an outage never slips into the aggregate disguised as a zero.

Three threats to validity deserve naming rather than leaving for a reader to find.

Construct validity comes first, and it lives in the judge. Faithfulness as measured is whatever the judge model scores as grounded, and Section 4.6 records an occasion when that instrument was wrong in a way that understated the system. Inspection caught it. The protocol did not, which is an uncomfortable thing to write and an honest one.

Internal validity is the second, and it lives in the gold answers. One annotator wrote them, and the same annotator checked them. Three came back wrong. A set yielding three errors to a single reviewer may well hold others that the same reviewer would not catch.

External validity is the third. Seven Gulf Cooperation Council companies, most of them banks, cannot stand in for a market, and twenty United States companies cannot stand in for a filing universe. Section 6.3 carries all three forward rather than letting the results read as broader than they are.

3.7 Ethical Considerations

Ethics came out simply here, because no human participants took part. Section 7.2 of the Thesis Handbook makes that decisive, and no formal Ethics Committee submission was required.

The work gathers no personally identifiable data either. Run it past the three categories the handbook flags for review and it is none of them: not a surveillance system, not an autonomous decision system, not a generative content system. Public corporate filings are

what it reads, out of the SEC EDGAR database and Bahrain Bourse. Under Bahrain Personal Data Protection Law, Law No. 30 of 2018, no personal data is being processed at all.

Responsible use, meanwhile, was built in at the start rather than bolted on at the end. Source document, section, confidence: every answer names all three. And every answer closes by reminding the reader it is not financial advice.

Chapter 4 Implementation

4.1 Development Environment

Development ran in Python on macOS, against the dependency set declared in the repository requirements file. The vector store, the sparse index, and the reranker stay on the local machine. Embeddings and generation are the exceptions and reach the Gemini API.

Streamlit drives the dashboard, and it calls the same code paths the evaluation calls. Nothing separate was built for demonstration. Had there been a demonstration path of its own, the interface could have drifted from the measured system without anything revealing the drift.

Where that local and remote boundary falls is also where exact reproducibility ends. Local computation is deterministic: the same query against the same index returns the same candidates in the same order, because BM25 scoring, the HNSW lookup, and the cross encoder are fixed operations over fixed data. The provider offers no equivalent guarantee. Identical prompts can produce different generations, and a call that succeeded once may time out on the next. Five runs rather than one follow directly from that asymmetry, and so does the pattern in Chapter 5, where retrieval figures hold to a tenth of a point and answer figures carry a visible standard deviation.

4.2 Data Pipeline

Filings arrive from SEC EDGAR. The pipeline parses each one into named sections, chunks every section recursively as Section 3.3.2 describes, embeds the chunks, and writes them into ChromaDB. Bahrain Bourse reports follow the same path through a dedicated loader.

Identifiers are where the design turned out to matter. Each encodes the company ticker, the filing type, the filing date, the section, and an index within that section, so scoping a query to one company reduces to a prefix test rather than a metadata lookup. That was convenient until Week 5, when the defect described in Section 4.6 showed what a prefix test costs if it is written carelessly. A ticker that prefixes a longer ticker will match it. JPM matches JPMX unless the delimiter is matched too, and a scoped search would then return another company's chunks with no error to announce it. Section 4.5 describes the regression test that closes this.

Isolated collections per corpus came from the same instinct. A Gulf Cooperation Council query cannot reach United States filings, and the reverse holds, because the two corpora never share an index. Duplication in the ingestion code is the price. The cross market claim in Chapter 5 rests on the two being measured independently, and a shared index carrying a

market field would have made that guarantee a matter of correct filtering rather than of structure. A third collection holds demonstration material and sits outside every evaluation path, so nothing prepared for a demonstration reaches a reported number.

4.3 Model Implementation

The agent follows the OpenAI Agents SDK ReAct pattern and reaches Gemini through LiteLLM. Its three tools are registered as function calling tools, and the agent selects among them from the question rather than sending every query through one general retrieval call. Each tool builds a query, embeds it, runs the hybrid search scoped to the target company, and formats what returns with source and confidence attached.

What the system prompt demands is narrower than it may appear. The agent answers from retrieved content only. Where a tool returns nothing, it says so rather than reaching for background knowledge.

That last requirement is enforced rather than merely requested, and the distinction carries into Chapter 5. A tool returning no passages leaves the agent with nothing to ground an answer in, so it abstains. The closed book model, given the same question, answers anyway. That is why the comparison between them is a comparison of behaviours and not only of accuracies.

Provider failures are held apart from wrong answers, deliberately. A call that times out or errors is retried with backoff; a question that still cannot be completed is recorded as an error, not scored as incorrect. Section 5.5 reports how many questions each run lost that way.

4.4 Indexing and Embedding Procedure

No model is trained here, so what stands in place of a training procedure is the embedding and indexing of the corpus. The 18,136 chunks of the United States corpus and the 845 chunks of the Gulf Cooperation Council corpus were embedded with gemini-embedding-001 at 1536 dimensions, then upserted into ChromaDB.

One constraint shaped how that ran. ChromaDB fixes a collection's vector dimension at creation, so the move from the earlier default embedding to the 1536 dimension Gemini embedding could not be made in place. The collection had to be deleted and rebuilt. Rebuilding 18,136 chunks against a metered API is not something to attempt twice, which is why the re embedding script checkpoints after every chunk and resumes where it stopped.

4.5 Software Engineering Practices

The repository carries a README, an MIT licence, and the evaluation result files, so the numbers in this thesis can be recomputed rather than accepted on faith. Work proceeded through version control on GitHub with feature branches merged through pull requests.

A suite of offline tests guards the contracts the evaluation rests on. Some are straightforward: a Gulf Cooperation Council ticker must route to the Bahrain Bourse collection, an empty prediction must score as a hallucination. Others exist because something went wrong. Four tests hold the company scoping in place, checking that the filter runs before ranking and not after, that a scoped search never leaks another company, that the unscoped path still ranks globally, and that one ticker cannot match another whose name it happens to prefix. That set was written directly against the defect in Section 4.6, and it fails under the original ordering.

All of it runs in continuous integration on every push, on two Python versions. A second job guards the evidence itself: it asserts that every result file cited in this thesis is present in the repository and that the headline figures still match what those files contain, so a reported number cannot quietly drift away from the data behind it. Bulk data, meaning the vector index and the raw filings, stays out of version control because it is large and re-downloadable; the result files are tracked precisely because they are the evidence.

4.6 Challenges and Solutions

Three challenges were significant enough to record, since each shaped the final system.

The first was a retrieval defect found in Week 5. While preparing a demonstration query it emerged that although all JPMorgan filings were indexed, the sparse retriever returned zero results for a JPMorgan query. Instrumenting the retrieval function showed that ticker filtering ran after the global top-k selection: a company own passages were discarded before the company filter applied, so if those passages were not in the global top-k they never reached the filtered result. The top matching JPMorgan passage ranked nineteenth globally for the example query, just beyond the threshold in use. Restricting the candidate pool to the target company before ranking corrected the defect and raised the FinanceBench retrieval hit rate from 54.8 percent to 96.8 percent. Because the candidate pool could no longer be silently truncated, the measured contribution of cross encoder reranking also rose, from 3.2 to 12.9 percentage points.

The second was compute access. Institutional cloud access did not materialise, so the codebase was migrated to the Gemini API on a billed tier. The embedding and generation

paths were rewritten around the Gemini SDK and LiteLLM, with retry and backoff on transient provider errors and a user safe error path that never leaks internal tracebacks.

The third was a measurement defect in the faithfulness judge. The judge truncated the retrieved context it saw to a fixed character cap that was smaller than the tool outputs for the Gulf Cooperation Council questions, which are long. The judge was therefore scoring faithfulness against a truncated view of the evidence, which understated faithfulness on that set. Raising the cap so that the judge sees the full evidence corrected the measurement. This is recorded as a methodological finding in its own right, since a faithfulness judge that cannot see the whole evidence will penalise a grounded answer unfairly.

One thing links all three, and it is worth drawing out. Not one was caught by the test suite. The retrieval defect surfaced while preparing a demonstration query, the compute problem announced itself as an absence rather than a failure, and the judge defect was found by reading what the judge was being passed rather than by any assertion about its output. Unit tests confirm that components behave as their authors expected; these were all cases where the expectation itself was wrong. Hence the regression tests described in Section 4.5, which were written afterwards, against behaviour observed in the system rather than behaviour imagined in advance.

4.7 Explainability and the Reasoning Trace

Delivery is through an interactive dashboard, shown below in its Ask Agent view. A board member types a question in natural language and receives a grounded, source attributed answer. The same interface carries a company filter, the retrieval controls, and live corpus and provider status, so the tool can be used without reading the code beneath it.

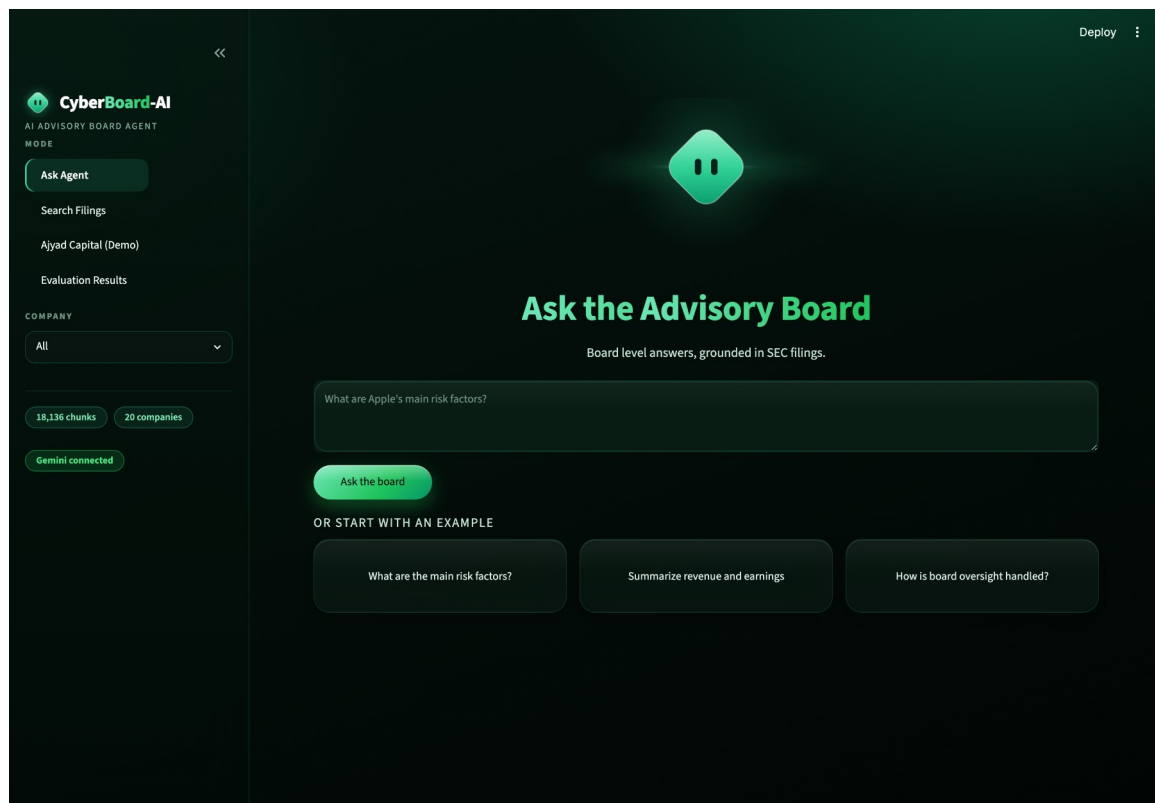


Figure 2. The CyberBoard-AI advisory dashboard in Ask Agent mode. A board member asks a question in natural language and the agent returns a grounded, source attributed answer.

Two mechanisms carry the explainability, and between them they address RQ3 and Objective O5. The first is attribution on every retrieved passage: company ticker, filing type, filing date, section, and a confidence score shown as a percentage. Those confidence values come from two different places. A reranked result reports a sigmoid transform of the cross encoder logit; a result served without reranking reports a normalised fusion score. Left as raw numbers the two would not be comparable, so both are mapped onto the same zero to one hundred scale and a board member reads one measure rather than two.

The second is a reasoning trace that opens the retrieval pipeline into three stages a reader can inspect. Candidate scoring reports how many sparse and dense candidates were found and lists the top BM25 matches with their scores. Hybrid fusion reports the rule applied, 0.7 times the dense score plus 0.3 times the sparse score with both normalised to the zero to one range, and the size of the pool passed forward. Cross encoder reranking names the model, ms-marco-MiniLM-L-6-v2, states whether the order changed, and tabulates the final ranks with their logits and confidence values. Nothing in that sequence is hidden. A board member can follow how a passage was found, fused and ranked, and weigh the answer accordingly.

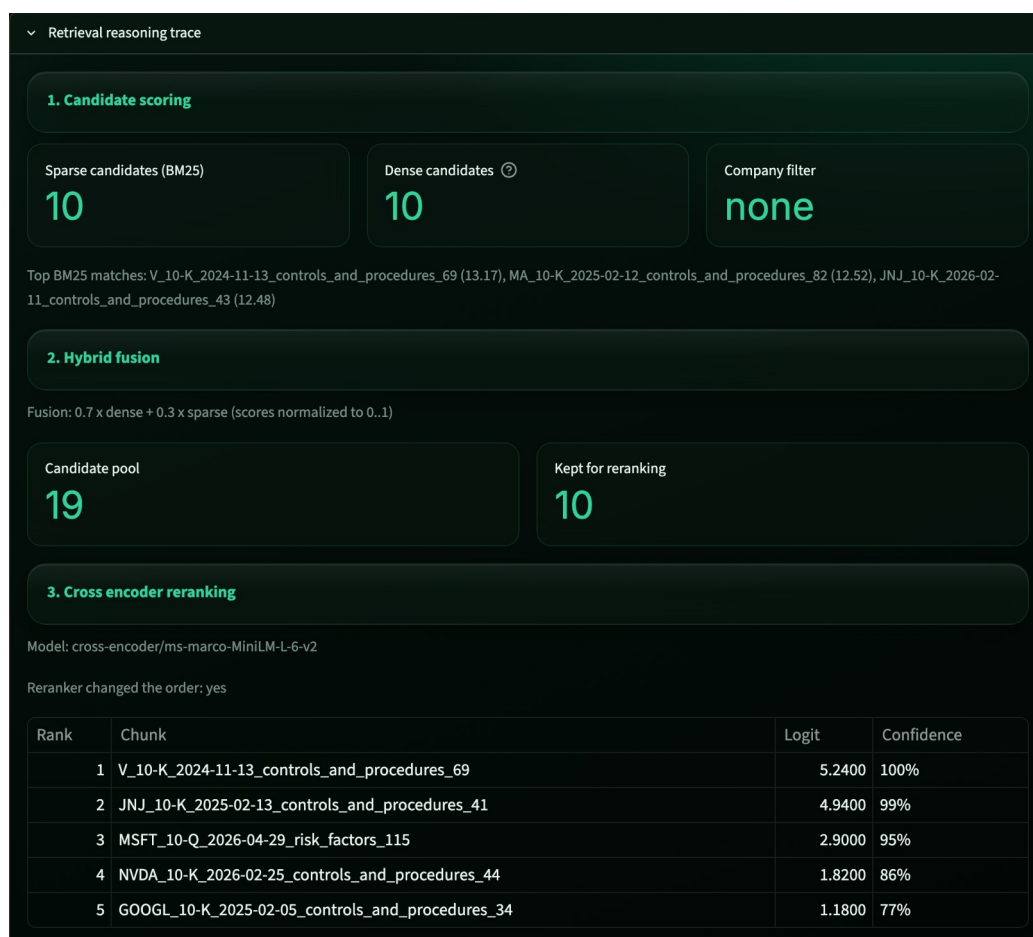


Figure 3. *The CyberBoard-AI reasoning trace for the query cybersecurity risk factors, showing candidate scoring, hybrid fusion, and cross encoder reranking, with source attribution and a confidence score on each result.*

An evaluation view sits alongside the per query trace, shown below. It renders the retrieval and ablation results reported in Chapter 5 directly from the stored result files, so the measured performance is visible in the same tool a board member uses to ask questions.



Figure 4. The evaluation view of the dashboard, rendering the FinanceBench retrieval hit rate and the per company breakdown from the stored result files.

Chapter 5 Results and Evaluation

5.1 Experimental Setup

Two corpora carry the results. The United States corpus, 18,136 chunks, is measured for retrieval quality against the FinanceBench matched questions.

The Gulf Cooperation Council corpus, made of Bahrain Bourse annual reports, is measured twice: once for retrieval quality, once for end to end answer quality, against a hand verified set of 35 questions. A further 19 numerical questions on recent United States filings serve as a second check on answer quality. Nothing in this chapter is typed in by hand. Every figure comes from a result file in the repository, recomputable with the commands in Appendix A.

The Gulf Cooperation Council question set is built to spread across the market rather than concentrate on one issuer. Seven Bahrain Bourse companies carry five questions each: National Bank of Bahrain, BBK, Aluminium Bahrain, Beyon, GFH Financial Group, Bahrain Islamic Bank, and Kuwait Finance House Bahrain. Banking dominates the Bahrain Bourse, so the set is weighted toward it, with an industrial issuer and a telecommunications group included so that the questions are not purely about banking disclosure. Each question was written against the company published annual report and each gold answer was checked back to the statement it came from.

5.2 Primary Results

Table 3 sets out end to end answer quality on the Gulf Cooperation Council set, the full agent beside both baselines. Averaged across five runs the agent reached 96.8 percent accuracy, standard deviation 0.1, and 99.4 percent faithfulness, standard deviation 1.2. Single pass retrieval managed 82.9 percent. Take retrieval away altogether and the closed book model reaches 11.4 percent.

That model's faithfulness of zero deserves a word, since it looks worse than it is. It is definitional. With no retrieved context there is nothing for an answer to be grounded in, so the judge marks every closed book answer ungrounded.

Configuration	Accuracy	Faithfulness	Hallucination	Mean latency
No retrieval (closed book)	11.4%	0.0%	100.0%	6.54 s
Single pass retrieval (naive RAG)	82.9%	97.1%	2.9%	6.56 s
Full agent (proposed)	96.8%	99.4%	0.6%	9.25 s

Table 3. End to end answer quality on the 35 question Gulf Cooperation Council set. The full agent figures are the mean over five runs; the two baselines are a single run on the same questions. Source: data/gcc_eval/multi_run_summary.json and the two baseline result files.

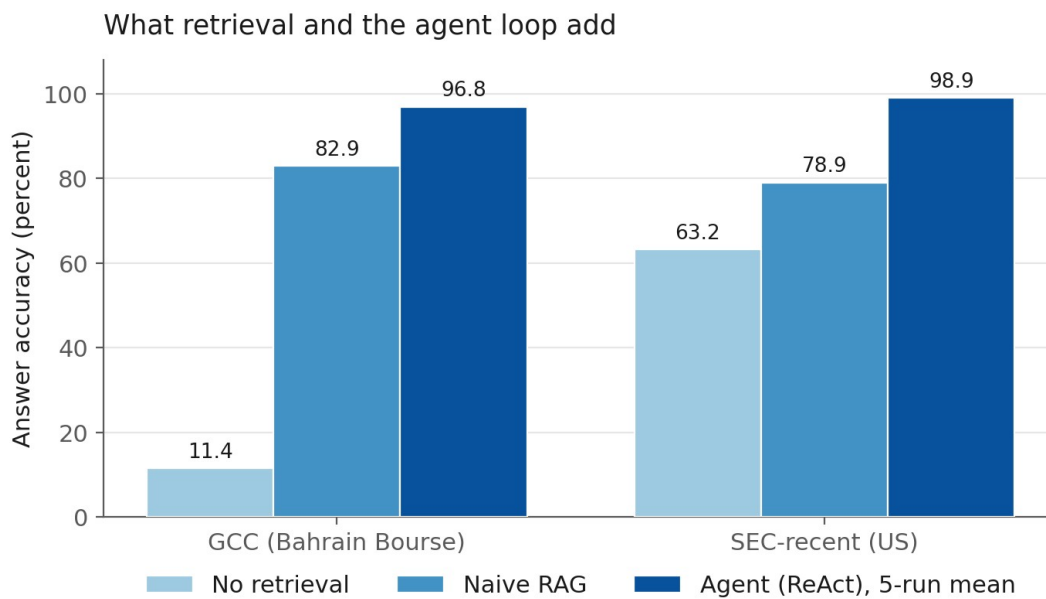


Figure 5. Answer accuracy across the three configurations on both corpora, the Gulf Cooperation Council set and the recent United States set. The agent bar is the mean over five runs, matching Table 3; the two baselines are a single run on the same questions. Source: the baseline result files and the multi run summaries.

The same agent was run on the 19 question set of recent United States numerical questions as an independent check that the result is not specific to one corpus. Over five runs on that set the agent reached 98.9 percent accuracy (standard deviation 2.1) and 96.8 percent faithfulness (standard deviation 4.2) over the scored questions, which indicates that the answer quality result transfers across the two markets.

The two baselines behave differently across the corpora in an informative way. The closed book model reaches 63.2 percent accuracy on the recent United States questions but only 11.4 percent on the Gulf Cooperation Council questions, because the financial figures of very large United States companies are more likely to reside in the model parametric memory than those of Bahrain Bourse companies. This widening gap is itself evidence that retrieval matters most precisely where the agent is intended to operate, on filings the model has not memorised.

5.3 Ablation Studies

On the United States corpus, Table 4 gives the retrieval ablation, and one component decides the outcome. Hybrid retrieval with reranking reaches 96.8 percent. Without reranking it reaches 83.9 percent. The gap, 12.9 percentage points, is what the reranking stage contributes. That 96.8 percent is the same figure that followed the ticker filtering fix of Section 4.6.

Configuration	Hit rate	Mean latency
Hybrid with rerank	96.8%	0.661 s
Hybrid without rerank	83.9%	0.049 s
Sparse only	96.8%	0.273 s
Alpha 0.5	96.8%	0.270 s
Alpha 0.9	96.8%	0.272 s

Table 4. Retrieval ablation on the United States corpus (FinanceBench). Source: data/financebench/ablation_results.json.

Table 5 turns to the Gulf Cooperation Council corpus, where the picture is more nuanced, and it is reported honestly rather than smoothed over. Hybrid retrieval without reranking hits 100 percent, a point above hybrid retrieval with reranking at 97.1 percent. Why? On a corpus this small and this homogeneous the cross encoder now and then reorders the single correct passage just below the cut. Reranking becomes a small net cost to recall instead of a gain. The finding is genuine, and it cuts against the earlier result: reranking helps markedly on the larger United States corpus and not on the Gulf Cooperation Council set.

Configuration	Hit rate	Mean latency
Hybrid without rerank	100.0%	0.797 s
Hybrid with rerank	97.1%	1.347 s
Dense only	97.1%	1.116 s
Alpha 0.5	97.1%	1.046 s
Alpha 0.9	97.1%	1.047 s
Sparse only	94.3%	0.222 s
Sparse without rerank	91.4%	0.001 s

Table 5. Retrieval ablation on the Gulf Cooperation Council corpus. Source: data/gcc_eval/ablation_results.json.

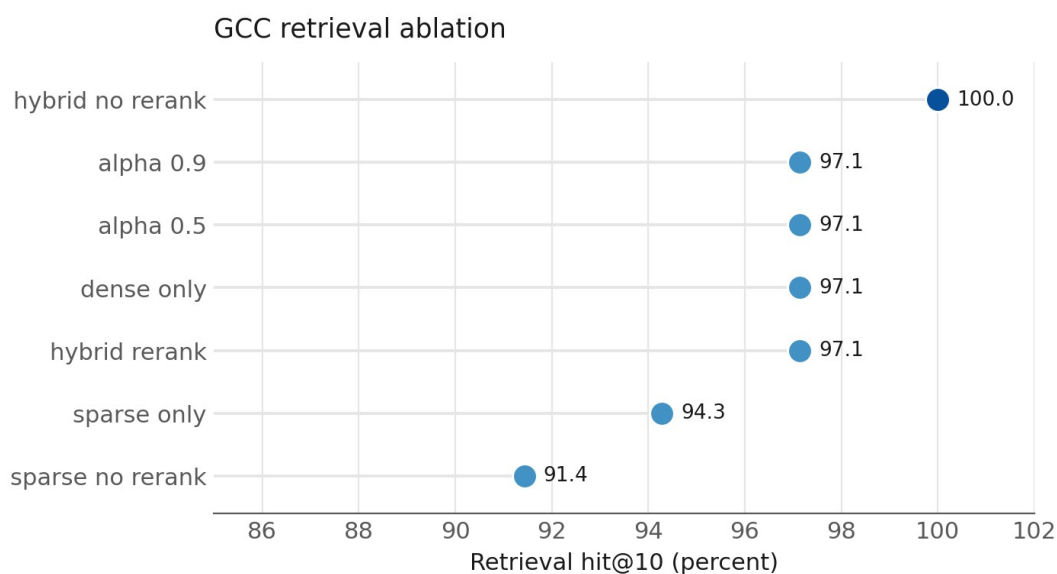


Figure 6. Retrieval hit rate by configuration on the Gulf Cooperation Council corpus.

5.4 Error Analysis

Where the closed book baseline fails, it fails predictably. It hands back figures that look plausible and rest on nothing, so the judge logs every one of its answers as ungrounded, and the four it gets right are the ones already sitting in parametric memory. Retrieval is not refining that baseline. It is the difference between answering and guessing.

Single pass retrieval is the more instructive failure, and the shape of it is easy to misread. Its accuracy is 82.9 percent, so six of the thirty five answers are wrong. Yet its faithfulness is 97.1 percent and only one answer in the set is logged as a hallucination. Grounding, in other words, did its job. The baseline is not inventing figures. It is failing to find the right ones.

That distinction matters more than the headline accuracy gap, so it is worth setting out case by case. Of the five questions where the agent was right and the single pass baseline wrong, two returned an explicit statement that the figure was not available in the retrieved context, which is a retrieval miss reported honestly rather than an error of reasoning. Two more were multi part questions: one asked for a chief executive and a board chair and returned the chief executive correctly while naming the wrong individual as chair, and one asked for owner equity and earnings per share and returned the equity figure correctly while reporting that the earnings per share was unavailable. Half a question answered is still a question wrong.

The fifth case is the one worth dwelling on. Asked for a total income figure, the baseline returned a number that appears in the company own report, cited it correctly, and was still wrong, because it had read the five year financial summary rather than the consolidated

income statement. Two tables in one document, both plausible, only one of them the answer to the question asked. No hallucination check catches that, since nothing was fabricated. The passage was real, the citation was real, and the source was the wrong one.

Each of those five failures is a retrieval problem wearing the costume of a reasoning problem, and that is precisely the gap the agentic layer addresses. A scoped tool call goes after the specific quantity a question turns on, and a second call can go after the second quantity when a question asks for two. Where the baseline gets one retrieval and must make do, the agent gets to ask again. Notice also that the five cases fall across five different companies, so the advantage is not an artefact of one issuer whose report happens to suit the agent.

On the retrieval side of the Gulf Cooperation Council corpus there is a single miss under reranking, and it is the reordering case from Section 5.3. The passage was retrieved. It simply landed below the cut.

5.5 Statistical Significance

Since the full agent and the single pass baseline see identical questions, a paired McNemar test is the honest way to compare them. On the primary evaluation run, 33 Gulf Cooperation Council questions were scored by both systems. Where the two disagreed, they disagreed in one direction only, five times the agent was right and the baseline wrong, and never once the other way round. The exact test on those five discordant pairs returns a two sided p value of 0.0625. Conventional practice wants 0.05, so this misses. The reason is not a feeble effect, though. It is arithmetic: five discordant pairs cannot reach significance on an exact test even when one side wins every single time. Direction, at least, is beyond doubt, and Section 7.3 sets out the fix, which is a bigger hand verified set.

How close is 0.0625 to the conventional threshold? Closer than the miss suggests. When every discordant pair falls the same way, the exact two sided p value is simply twice one half raised to the number of pairs, so five pairs give 0.0625 and six would give 0.03125. One further question on which the agent succeeded and the baseline failed would have carried this result across the 0.05 line, with nothing else about the system changed. That is a statement about the size of the evaluation set rather than the strength of the effect, and it is the sharpest argument available for the larger gold set proposed in Section 7.3.

Repeating the Gulf Cooperation Council evaluation five times gave a picture of variance. No run had to be discarded, though none came through entirely clean either. Transient provider errors cost three, four, two, five, and five questions across the five runs, so between 30 and 33

of the 35 were scored on any given pass. Those lost questions were set aside for that run rather than marked wrong, on the reasoning that a passing outage should not masquerade as a failure. Accuracy averaged 96.8 percent, with a standard deviation of 0.1. Faithfulness averaged 99.4 percent, standard deviation 1.2. It is this five run mean, not any single run, that the abstract and Section 5.2 quote, simply because it is the sturdier number. The primary run behind the McNemar test happened to reach 100 percent, which sits at the top of the range.

5.6 Computational Cost and Environmental Impact

Cost is easy to characterise, because nothing is trained. It comes down to embedding the corpus once, then paying for inference per query. The 18,136 chunks are embedded a single time and never again. From then on retrieval happens locally, and a question costs a few Gemini calls against a flash class model. The latency figures show what the agent buys and what it charges: 9.25 seconds mean end to end for the full agent, roughly 6.5 seconds for the single pass and closed book paths. That difference is reasoning and tool calls. Taken together the footprint is small, one workstation carrying the retrieval infrastructure and a short, sparse set of language model calls on top.

The mean latency figure hides a skew that a board deploying this would feel. Across the primary Gulf Cooperation Council run the agent averaged 9.25 seconds per question, but the median was 7.00 seconds and the slowest single question took 22.13. Most questions resolve in one tool call and finish quickly; a minority send the agent back for a second or third call, and those are the ones that stretch. So the typical wait is noticeably shorter than the average suggests, while the worst case is more than three times the median. Reporting the mean alone would flatter the system in the common case and understate it in the tail.

Chapter 6 Discussion

6.1 Interpretation of Findings

Across all three research questions the findings point one way. Retrieval, once the ticker filtering defect is gone, is strong: a 96.8 percent hit rate on the United States corpus and 97.1 percent on the Gulf Cooperation Council corpus. That is RQ1. The agent then beats single pass retrieval on answers, 96.8 percent accuracy against 82.9 percent, which is RQ2, carrying the caveat about statistical power set out in Section 5.5. As for RQ3, it is the explainability layer that settles it, since source attribution, confidence scores, and the reasoning trace together turn an answer from something opaque into something a reader can check.

One result refuses to point that way, and it surfaced only because the ablation was run twice, once per corpus. Reranking lifts the United States hit rate by 12.9 percentage points and costs the Gulf Cooperation Council corpus 2.9. Size explains part of that. Homogeneity explains more. The United States corpus holds 18,136 chunks from twenty companies across several industries, so a query meets many plausible near matches and the reranker earns its keep discriminating among them. The Gulf Cooperation Council corpus holds 845 chunks from seven companies, five of them banks filing structurally similar statements. There the correct passage already sits at the top of the fused ranking more often than not, and a reranker asked to improve on a correct ordering has nothing to win and one place to lose.

Read that way, the finding is not a defect in the reranker but a boundary condition on it. Cross encoder reranking pays when the candidate pool is large and genuinely ambiguous. It stops paying, and turns slightly negative, when the pool is small and uniform. A system serving both markets should therefore treat reranking as a per corpus decision rather than a global setting, which is a concrete design consequence this evaluation produced and the literature reviewed in Section 2.5 anticipated only in the abstract.

6.2 Comparison with Prior Work

Set beside prior work, the retrieval hit rate falls within the range financial retrieval systems such as FinSage [3] report. That matters, because it supports the claim that this hybrid design is genuinely competitive rather than merely functional. What the agent adds over single pass retrieval also lines up with the argument threaded through the ReAct line of work [7], namely that interleaving reasoning with tool calls lifts task accuracy. The Gulf Cooperation Council ablation cuts the other way, and usefully so. Reranking there is a small net cost, which

unsettles the common assumption that a cross encoder reranker always helps. On the larger, more heterogeneous corpus it helps. On the smaller one it does not.

The closed book model supplies a comparison the prior work does not, because it was run on both corpora rather than one. It answers 63.2 percent of the 19 recent United States questions from parametric memory alone and 11.4 percent of the 35 Gulf Cooperation Council questions. That is not a small difference in degree. It is a fivefold gap, and it says something about whose disclosure a general purpose model has absorbed. Filings from very large United States issuers are heavily represented in the text such models are trained on; Bahrain Bourse annual reports are not.

Two things follow. The first is methodological: any evaluation of a retrieval system that uses only United States data will understate what retrieval contributes, because the model can partly answer from memory and the baseline looks stronger than it is. The second is broader. If model memory is thin for a market, then retrieval grounding is not a refinement for that market but the whole of the capability, and the case for building such systems is strongest precisely where the training data is weakest. That argument is available here only because the evaluation spans two markets.

6.3 Limitations

Several limitations bound the strength of the claims.

- RQ3 is addressed by design rather than by measurement. Source attribution, confidence scores, and the reasoning trace were built and are demonstrably present in every answer, but their effect on how a board member perceives transparency and verifiability was not measured with users. Demonstrating that a system exposes its evidence is not the same as demonstrating that the exposure changes what a reader can verify, and Section 7.3 sets out the study that would close the difference.
- The Gulf Cooperation Council gold set was written and checked by a single annotator, the author, so no inter-annotator agreement can be reported for it. Three gold answers proved wrong during that check and were corrected, which is evidence that the instrument is fallible and that a second independent annotator would put the ground truth on firmer footing.
- With 35 questions, the Gulf Cooperation Council evaluation set limits statistical power, as Section 5.5 makes explicit.

- FinanceBench questions cover 2016 to 2023 while the corpus covers 2024 to 2026, so FinanceBench serves retrieval quality through evidence overlap and not answer accuracy against its own gold answers.
- Weighted as it is toward large listed companies, the corpus may not transfer to small or unlisted firms.
- Faithfulness rests on an LLM judge, itself a model and therefore capable of error; the judge was corrected once during this study, when a context truncation defect understated faithfulness.

6.4 Responsible AI Reflection

Responsible use here was designed in, not bolted on afterward. Each answer cites the document, the section, and a confidence, and each one carries a disclaimer that the output is not financial advice. Where its tools return no evidence the system abstains instead of reaching into parametric memory, and that behaviour is precisely what separates the full agent from the closed book baseline. In a governance setting it is the behaviour that matters most. The data themselves are public disclosures, so no personal data is processed. One concern does remain, which is representational bias toward large listed companies. The Gulf Cooperation Council set softens it without removing it.

One aspect of the interface deserves a frank caveat, because it could mislead exactly the reader it is meant to serve. The confidence percentage shown beside each passage is a retrieval confidence, not an answer confidence. For a reranked result it is a sigmoid transform of the cross encoder logit, and for an unreranked one a normalised fusion score, so what it expresses is how well that passage matches the query. It does not express the probability that the answer built on top of it is correct. A board member glancing at a figure of ninety nine percent could reasonably read it as the latter. The number is honest about what it measures and silent about what it does not, and that silence is a design weakness worth naming rather than defending.

Automation bias is the related risk, and explainability can worsen it rather than cure it. A system that displays sources, scores, and a reasoning trace looks rigorous, and looking rigorous encourages trust that the underlying accuracy may not have earned. Fritz-Morgenthal et al. [27] and the CFA Institute report [28] both press the point that an explanation serves the reader only if the reader can actually evaluate it. The mitigation adopted here is modest and partial: the system abstains when its tools return nothing, states

that its output is not financial advice, and shows the retrieved passage itself so a board member can read the source rather than take the summary on trust. Whether that is sufficient is precisely what the user study in Section 7.3 would establish, and it is not established here.

On data governance the position is straightforward and worth stating plainly. Everything indexed is public corporate disclosure, published by the issuers themselves through SEC EDGAR and Bahrain Bourse, so no personal data is processed and Bahrain Personal Data Protection Law imposes no obligation that bites on this corpus. Queries are a different matter in a deployed setting. A board member's questions reveal what that board is examining, and a production deployment sending those queries to a third party model provider would be exporting a signal about the institution's own concerns. Nothing in this thesis resolves that; a deployment would need to.

6.5 Practical Implications

What does this mean for a board? Chiefly that an advisory agent can shorten the path from a governance question to the cited passage in a filing, while keeping both the evidence and the system's confidence in view, so the board member stays the decision maker. The cross market result carries a second implication. An approach developed against United States filings can move across to Gulf Cooperation Council filings without redesign, which matters for regional institutions obliged to read both.

Cost sits low enough not to be the obstacle. Nothing is trained, the corpus is embedded once, and retrieval runs locally thereafter, so the recurring expense is a handful of model calls for each question asked. An institution already paying for analyst time to locate figures in filings is not going to find the inference bill decisive. What it will find decisive is whether the index is current.

That is the real operational burden, and it is easy to underestimate. A filing corpus goes stale on a fixed schedule, quarterly and annually, and an advisory agent answering from last year's annual report while presenting a confident citation is worse than no tool at all, because the citation makes the staleness invisible. Any deployment therefore needs an ingestion process tied to the reporting calendar and, ideally, an answer that carries the filing date prominently enough that a reader notices when it is old. The system does surface the filing date with every passage, which helps, but the discipline has to live in the operating process rather than in the interface alone.

A last implication concerns where a tool like this belongs in a board's workflow. It is a retrieval and summarisation aid, and the evaluation supports it as that. It is not a judgement engine, it has not been tested against the questions that actually divide boards, and the disclaimer it prints on every answer is not decoration. The honest claim is narrower than the technology invites: the agent shortens the distance between a governance question and the disclosure that speaks to it, and leaves the governing to the people charged with it.

One last practical point, and it is about who owns the output. An answer this system produces is evidence a board member has read, not a position the board has taken. The distinction sounds pedantic until a decision is questioned afterwards, at which point it matters a great deal whether the reasoning on record belongs to the directors or to a retrieval pipeline they consulted. Keeping the tool on the evidence side of that line is a governance choice as much as a design one.

Chapter 7 Conclusion and Future Work

7.1 Summary of Contributions

This thesis built and evaluated CyberBoard-AI, an advisory board agent unifying hybrid retrieval, agentic tool use, and explainability for corporate governance advisory, and it tested that agent across a United States benchmark and Gulf Cooperation Council filings. Along the way it documented and corrected a retrieval defect, which lifted the FinanceBench hit rate from 54.8 to 96.8 percent, and it reported every result from a reproducible result file. It also established, by running the closed book baseline on both corpora, that a general purpose model answers 63.2 percent of the recent United States questions from parametric memory and 11.4 percent of the Gulf Cooperation Council questions, which means retrieval grounding matters most where a market is thinly represented in training data.

Two findings sit alongside those contributions and were not anticipated when the work began. Cross encoder reranking, treated in much of the literature as a component that reliably helps, helped substantially on the larger United States corpus and cost a little recall on the smaller and more homogeneous Gulf Cooperation Council one, which makes it a per corpus decision rather than a default. And a faithfulness judge, left unexamined, misreported the very metric it existed to measure, until the truncation defect described in Section 4.6 was found and corrected. Both are reported here because a negative or awkward result that survives scrutiny is worth more than a clean one that has not been scrutinised.

7.2 Answers to the Research Questions

RQ1 is answered in the affirmative: the agentic retrieval system retrieves and combines filing information at 96.8 and 97.1 percent hit rate across the two corpora, and it answers faithfully. RQ2 is answered in the affirmative as to direction, the full agent reaching 96.8 percent accuracy against 82.9 percent for single pass retrieval, subject to the sample size caveat on statistical significance. RQ3 is answered through the source attribution, confidence scores, and reasoning trace that leave each advisory output verifiable.

RQ3 deserves a more careful answer than the other two, because the honest one is partial. The features it asks about were built, and they are present in every answer the system returns: the source document, the section, the filing date, a confidence value, and a reasoning trace that opens the retrieval pipeline into its three stages. What was not done is the measurement. No board member was asked whether any of it changed what they could verify or how far

they trusted the result. The affirmative part of the answer concerns what the system exposes, and it is evidenced. The part concerning effect on a reader is asserted, and Section 7.3 sets out the study that would settle it.

7.3 Future Work

Four directions would strengthen and extend the work.

- Expand the hand verified Gulf Cooperation Council evaluation set beyond 35 questions so that the McNemar comparison reaches conventional significance.
- Run an explainability study with the intended readers. Put source attribution, confidence scores, and the reasoning trace in front of board members and governance staff, and measure whether those features change what a reader can actually verify and how much they trust the result. That is the evidence RQ3 asks for and this thesis does not supply.
- Extend the corpus toward small and unlisted companies, and toward additional Gulf Cooperation Council markets, testing the representational limits noted in Section 6.3.
- Explore a routed multi agent design, in the direction of the work surveyed in Section 2.4, and measure whether routing adds value over the single agent with three tools used here.

Were only one of the four to be attempted, it should be the second. The first, third, and fourth would each extend what the system does or how widely it has been tested, and all three are ordinary research increments. The explainability study is different in kind, because it addresses the one research question this thesis answers by construction rather than by evidence. Until a board member has sat with the reasoning trace and been asked what they could actually verify from it, the central claim of the explainability layer remains a designer's belief about a reader rather than a finding about one. It is also the cheapest of the four to run, needing a modest number of participants and no additional engineering, which makes its absence here a matter of the thirteen week scope rather than of difficulty.

7.4 Concluding Remarks

A board member does not need a system that sounds authoritative. What is needed is an answer traceable to the filing it came from, with a stated confidence and a clear boundary around what the system knows. CyberBoard-AI is a step toward that standard. Its most valuable single result, tellingly, was not a feature delivered on schedule. It was a defect, found by running a real query against a real company and noticing that the answer was wrong.

That is the note to end on, because it is the habit the whole project rests upon. The retrieval defect was not caught by a test suite or a metric; it was caught by looking at one answer and finding it implausible. The faithfulness judge was not corrected by the protocol that ran it, but by reading what the judge was actually being shown. A system that surfaces its evidence is valuable in the end for the same reason: it gives a reader something to check against, and preserves the possibility of noticing that a confident answer is wrong.

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Appendix A Source Code Repository and Reproduction

The repository holds the source, the MIT licence, and every result file Chapter 5 draws on:
<https://github.com/hatemmsc-ops/CyberBoard-AI>

Chapter 5 belongs to one commit. Cite that, not the branch, since the branch will move:

```
Commit 9660fb81282a491d96f404a50422adfde8fb7eb9 (short 9660fb8)
```

Inspect it at the address below, or pin a working copy to it with the command beneath:

```
https://github.com/hatemmsc-ops/CyberBoard-AI/commit/  
9660fb81282a491d96f404a50422adfde8fb7eb9  
git checkout 9660fb81282a491d96f404a50422adfde8fb7eb9
```

From the repository root, five commands rebuild the numbers:

```
python -m src.evaluation.financebench_eval      # United States retrieval hit  
rate  
python -m src.evaluation.ablation_study         # retrieval ablation  
python -m src.evaluation.full_eval --set gcc    # Gulf Cooperation Council  
answer quality  
python -m src.evaluation.baselines --set gcc    # no retrieval and single  
pass baselines  
python -m src.evaluation.multi_run --set gcc    # repeated runs for variance
```

Appendix B Dataset Details

Item	Value	Source
SEC filings parsed	116 (20 companies)	data/sec_filings/manifest.csv
Chunks indexed (US corpus)	18,136	ChromaDB collection
Chunk size / overlap	512 tokens / 64 tokens	config.py
FinanceBench matched questions	31	data/financebench/matched_questions.json
Gulf Cooperation Council questions	35 (hand verified)	data/gcc_eval/questions.json
Recent US numerical questions	19 (hand verified)	data/sec_recent/questions.json

Table 6. Dataset summary.

Appendix C Hyperparameters

Parameter	Value
Sparse retriever	BM25 (rank-bm25)
Dense embedding	gemini-embedding-001, 1536 dimensions
Fusion weight alpha	0.7 (dense favoured)
Reranker	cross encoder ms-marco-MiniLM-L-6-v2
Generation model	Gemini flash, temperature 0.1
Judge model	Gemini, temperature 0
Retrieval hit threshold	30% gold evidence word overlap

Table 7. Key hyperparameters.

Appendix D Ethics Approval

Ethics approval: not required, for reasons Section 3.7 sets out. No human participants took part and no personal data is processed. Section 7.2 of the Thesis Handbook flags three categories for review, those being surveillance, autonomous decision making, and generative content. This study sits in none of them.

Appendix E Supervisor Meeting Log

Supervision was weekly. Dr. Joshua Samuel and the author met on schedule through the thesis period, with a written progress update sent ahead of each meeting and minutes circulated afterwards. Eight meeting records are held. The first is dated 2 July 2026, and the sequence runs weekly from there through week 11 of the programme. Each record names the attendees, the results reviewed, the decisions taken, and the actions agreed with their owners and due dates.

The full log is open to view at the address below.

<https://docs.google.com/document/d/1rBB9Yf7VZ7d2vK4Fa6lMSVv93fg5v6s00-BpN4l6uSQ/edit>